🔑 **Answer Key — Teacher Copy · Lesson 8-1**

MSTAR PRACTICE WORKSHEET

# Lesson 8-1: Understand Equations and Their Solutions

Grade 6 Mathematics · Standard 6.AT.8

*These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.*

## Part A — Skills Check

One point each.

**Question 1** SELECTED RESPONSE — CORRECT: A

Which equation represents: 'A number plus 12 equals 30'?

- **A** n + 12 = 30
- **B** n – 12 = 30
- **C** 12n = 30
- **D** n / 12 = 30

'Plus' means addition, so the equation is n + 12 = 30.

**Question 2** SELECTED RESPONSE — CORRECT: A

Which equation represents: 'Seven less than a number is 18'?

- **A** n – 7 = 18
- **B** 7 – n = 18
- **C** n + 7 = 18
- **D** n / 7 = 18

'Seven less than a number' means start with the number and subtract 7: n – 7 = 18.

**Question 3** SELECTED RESPONSE — CORRECT: A

Which equation represents: 'Three times a number equals 21'?

- (A) 3n = 21
- (B) n + 3 = 21
- (C) n − 3 = 21
- (D) n / 3 = 21

'Three times' means multiply by 3, so the equation is 3n = 21.

**Question 4** SELECTED RESPONSE — CORRECT: B

Is n = 3 a solution of n + 5 = 9?

- (A) Yes, because 3 is smaller than 9
- (B) No, because 3 + 5 = 8, which does not equal 9
- (C) Yes, because 3 + 5 is close to 9
- (D) You cannot tell without solving first

Substitute the value and check whether the statement is true: 3 + 5 = 8, and 8 ≠ 9, so 3 is not a solution.

**Question 5** SELECTED RESPONSE — CORRECT: A

Which equation represents: 'A number divided by 4 equals 8'?

- (A) n / 4 = 8
- (B) 4n = 8
- (C) n − 4 = 8
- (D) n + 4 = 8

'Divided by 4' means division: n / 4 = 8.

**Question 6** SELECTED RESPONSE — CORRECT: A

A detective found some fingerprints on Monday and 14 more on Tuesday, for a total of 30. Which equation represents this?

- **A** f + 14 = 30
- **B** f – 14 = 30
- **C** 14f = 30
- **D** f / 14 = 30

The detective's Monday fingerprints (f) plus 14 more equals 30: f + 14 = 30.

## Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

**Question 7** TWO-PART QUESTION

**Part A — correct answer: B**

Marisol tests values in the equation 7 + m = 23. Which value of m makes the equation true?

A. m = 9

B. m = 16

C. m = 23

D. m = 30

Substitute each value for the variable. When m = 16, the left expression is 7 + 16 = 23 and the right expression is 23, so both sides balance. That makes 16 the solution.

- **A:** Substituting 9 gives 7 plus 9, which reaches only 16.
- **C:** That is the total, not the missing addend. Seven plus 23 comes to 30.
- **D:** That adds 7 and 23. To undo an addition of 7, subtract it from 23.

**Part B — correct answer: C**

Which reasoning best justifies that answer?

A. Adding the two numbers written in the equation, 7 + 23, gives the value of m.

B. The equal sign means the answer is always the number written to the right of it.

C. Substituting 16 for m gives 7 + 16 = 23, so both sides of the equation have the same value.

D. Any number can replace m because m is a variable, and variables can change.

A value is a solution only when substituting it makes the left expression equal the right expression. Substituting 16 gives 23 = 23, so the equation is true. Adding 7 and 23 gives 30, and 7 + 30 = 37, which does not balance.

**Question 8** SELECT ALL THAT APPLY — CORRECT: A, B, C

Select ALL statements that are true about equations and their solutions.

A. An equation is a math sentence that uses an equal sign to show two expressions have the same value.

B. You can test a value by substituting it for the variable and checking whether both sides are equal.

C. The value n = 6 is a solution of n + 9 = 15 because 6 + 9 = 15.

D. The expression 4y + 2 is an equation because it contains a variable.

E. The value x = 15 is a solution of x − 4 = 9 because 15 − 4 = 9.

The first three statements are true. The fourth is false because 4y + 2 has no equal sign, so it is an expression, not an equation. The fifth is false because 15 − 4 = 11, not 9; the value that makes x − 4 = 9 true is x = 13, since 13 − 4 = 9.

## Part C — Show Your Thinking

Model answer and scoring guidance.

**Question 9** WRITTEN RESPONSE · 2 POINTS

Write a real-world situation that can be modeled by the equation n / 5 = 9. Explain what n represents and verify your equation makes sense.

**Model answer:** The equation n / 5 = 9 describes one amount being shared into 5 equal groups with 9 in each group, so a matching story could be n stickers split evenly among 5 friends with each friend getting 9. Since 5 groups of 9 rebuild the whole, n = 9 × 5 = 45, and 45 ÷ 5 = 9 checks the story.

*Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.*

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